

Final Examination
Introduction to Massive Data Analysis Dec. 4, 2019

1. (12%) Yes/no questions. You don't have to justify the answers.
 - (a) PCY can handle some data sets without thrashing during the first pass, while A-Priori would run out of main memory and thrash.
 - (b) One way to measure the similarity of nodes in a graph with several types of nodes is to start a random walker at one node and allow it to wander, with a fixed probability of restarting at every node.
 - (c) The key to understanding what SVD offers is in viewing the r columns of U , Σ , and V as representing concepts that are hidden in the original matrix M .
 - (d) PageRank π is the stationary distribution of the Markov chain M i.e. $M\pi = \pi$ and $\sum_i \pi_i = 1$.
2. (8%) Prove that $\Pr[h_\pi(C_1) = h_\pi(C_2)] = \text{sim}(C_1, C_2)$, the probability that the minhash function for a random permutation π of rows produces the same value for two sets equals the Jaccard similarity of those sets.
3. (8%) Consider a Web site gathering statistics on how many unique users it has seen in each given month. The universal set is the set of logins for that site, and a stream element is generated each time someone logs in. The Flajolet-Martin algorithm can solve this problem if we do not have enough space to maintain the set of elements seen so far. Briefly describe the Flajolet-Martin algorithm and show how it works.
4. (4%) In the block-stripe parallel implementation of PageRank, show that the computational cost per iteration of Power method is $|M|(1+\epsilon) + (k+1)|r|$, where k is the number of block.
5. (10%) In the general case, the worst competitive ratio of the BALANCE (many bidders) on-line algorithm is $1-1/e \approx 0.63$.
 - (a) (3%) Gives the worst case example with N advertisers of the same budget B , and NB queries in N rounds.
 - (b) (7%) Show that the revenue of BALANCE is $BN(1-1/e)$, and thus the competitive ratio is $1-1/e$.
6. (8%) Here is a collection of twelve baskets. Each contains three of the six items 1 through 6.

$\{1, 2, 3\} \{2, 3, 4\} \{3, 4, 5\} \{4, 5, 6\}$
 $\{1, 3, 5\} \{2, 4, 6\} \{1, 3, 4\} \{2, 4, 5\}$
 $\{3, 5, 6\} \{1, 2, 4\} \{2, 3, 5\} \{3, 4, 6\}$

Suppose the support threshold is 4. On the first pass of the PCY Algorithm we use

a hash table with 11 buckets, and the set $\{i, j\}$ is hashed to bucket $i \times j \bmod 11$.

(a) By any method, compute the support for each item and each pair of items.

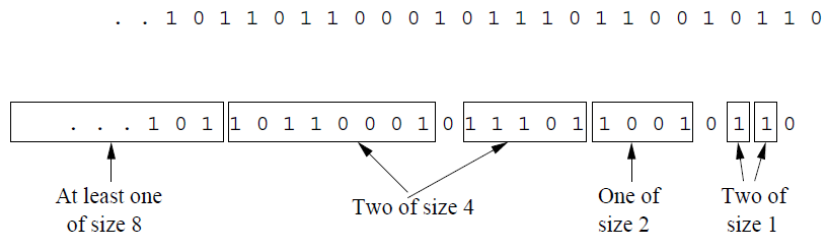
(b) Which pairs hash to which buckets?

(c) Which buckets are frequent?

(d) Which pairs are counted on the second pass of the PCY Algorithm?

7. (8%) In the demonstrative example of LSH for Min-Hash, we divide matrix M into 20 bands of 5 rows. If we had only 15 bands of 5 rows, show that the number of false positives would go down, but the number of false negatives would go up.

8. (10%) A demonstrated example of the simple DGIM algorithm is shown in the following figure. Consider $k = 10$. Then the query asks for the number of 1's in the ten rightmost bits, which happen to be 0110010110. Let the current timestamp (time of the rightmost bit) be t . Then the two buckets with one 1, having timestamps $t - 1$ and $t - 2$ are completely included in the answer. However, the rightmost bucket of size 4, with timestamp $t - 8$ is only partly included. DGIM estimate of the number of 1's in the last ten positions is thus 6. Of course the correct answer is 5.



(a) (5%) Show that DGIM requires $O(\log^2 N)$ bits to represent a window of N bits.

(b) (5%) The estimation error of the simple DGIM is no more than 50%, why?

9. (8%) A Bloom filter consists of (i) an array B of n bits, initially all 0's; (ii) a collection of k hash functions $h_1, h_2, \dots, h_k$. Each hash function maps "key" values to n buckets, corresponding to the n bits of B ; (iii) a set S of m key values. The purpose of the Bloom filter is to allow through all stream elements whose keys are in S , while rejecting most of the stream elements whose keys are not in S .

(a) (2%) Why the Bloom filters guarantee no false negatives?

(b) (6%) Calculate the probability of *false positive rate* of Bloom Filter with m keys, n bits, and k hash functions, and determine the optimal k .

10. (8%) A market-basket data set has 1 million baskets, each of which has 10 items that are chosen from a set of 100,000 different items (not all of which necessarily appear in even one basket). Suppose the support threshold is 100.

- (a) (4%) What are the minimum and maximum numbers of frequent items?
Explain your reasoning.
- (b) (4%) What are the minimum and maximum numbers of frequent pairs?
Explain your reasoning.
11. (8%) Suppose we apply the BALANCE algorithm with bids of 0 or 1 only, to a situation where advertiser A bids on query words x and y, while advertiser B bids on query words x and z. Both have a budget of \$2. Identify the sequences of queries that will certainly be handled optimally by the algorithm, and provide a one line explanation.
- (a) (2%) yzyy (b) (2%) xyzx (c) (2%) xyyy (d) (2%) yxzx
12. (10%)

	MMDS	Machine Learning	Data Structures	Dynamics	Mechanics
Diane	1	1	1	0	0
Ethan	2	2	2	0	0
Frank	1	1	1	0	0
Grace	5	5	5	0	0
Hank	0	0	0	2	2
Ingrid	0	0	0	3	3
Joe	0	0	0	1	1

Matrix M represents the ratings of Engineering courses taken by Stanford students. Each row of M represents the given student's set of ratings and the columns of M represent the classes. The labels for students and classes are provided along each row and column. An entry M_{ij} in M represents the student i 's rating for class j . Now, the SVD decomposition of matrix M is found to be as follows:

$$USV^T = \begin{bmatrix} .18 & 0 \\ .36 & 0 \\ .18 & 0 \\ .90 & 0 \\ 0 & .53 \\ 0 & .80 \\ 0 & .27 \end{bmatrix} \begin{bmatrix} 9.64 & 0 \\ 0 & 5.29 \end{bmatrix} \begin{bmatrix} .58 & .58 & .58 & 0 & 0 \\ 0 & 0 & 0 & .71 & .71 \end{bmatrix}$$

It's pretty clear that there are two concepts of classes here: the first three are Computer Science classes while the last two are Mechanical Engineering classes.

Suppose a new student named Tony has the following reviews: 4 for Machine Learning, 5 for Data Structures, and 2 for Dynamics. This gives a representation of Tony in the class space as $[0 \ 4 \ 5 \ 2 \ 0]$.

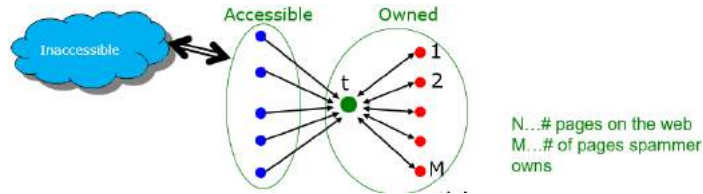
- (a) (5%) What is the representation of Tony in the concept space?
- (b) (5%) Explain what this representation predicts about how much Tony would like the two remaining classes that he has not reviewed (MMDS and Mechanics).

13. (6%) Suppose we are given a matrix $\mathbf{R}$ of recommendations. The element R_{iu} of this matrix corresponds to the rating given by user u to item i . The size of $\mathbf{R}$ is $m \times n$, where m is the number of movies, and n the numbers of users. Most of the elements of the matrix are unknown because each user can only rate a few movies. Our goal is to find two matrices $\mathbf{P}$ and $\mathbf{Q}$, such that $\mathbf{R} \approx \mathbf{Q}\mathbf{P}^T$. The dimensions of $\mathbf{Q}$ are $m \times k$, and the dimensions of $\mathbf{P}$ are $n \times k$. k is a parameter of the algorithm. We define the error as

$$E = \sum_{(i,u) \in \text{ratings}} (R_{iu} - q_i \cdot p_u^T)^2 + \lambda \left[\sum_u \|p_u\|_2^2 + \sum_i \|q_i\|_2^2 \right].$$

Let ε_{iu} denote the derivative of the error E with respect to R_{iu} . What is the expression for ε_{iu} ? What are the update equations for q_i and p_u in the Stochastic Gradient Decent Algorithm?

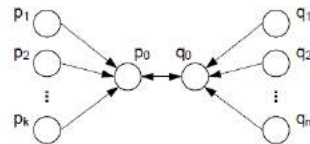
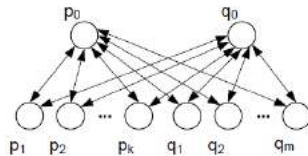
14. (12%) One of the most common and effective organizations for a link farm is shown below:



- (a) (4%) Let $\mathbf{x}$ denote the PageRank contributed by accessible pages and $\mathbf{y}$ be the PageRank of target page t . And rank of each “farm” page is $\frac{\beta y}{M} + \frac{1-\beta}{N}$. Show

that $y = \frac{\beta y}{1-\beta^2} + c \frac{M}{N}$, where $c = \frac{\beta}{1-\beta}$.

- (b) (8%) Suppose two spam farmers agree to link their spam farms. How would you link the pages in order to increase as much as possible the PageRank of each spam farm’s target page? One way of doing it is just having all boosting pages point to both of the targets, as presented in left figure. Another way of connecting the two target pages and removing all links, as shown in the right figure. Which one performs better? Explain your reason with some calculations as stated by (a).



國立清華大學試卷

記		分			
1	✓	2	✓		
3	✓	4	✓		
5	✓	6	✓		
7	✓	8	✓		
9	✓	10	✓		
11	✓	12	✓		
13	✓	14	✓		
15		16			
17		18			
19		20			
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科 目 巨資考古 2019

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日 期 _____

1. (a) X

(b) $\cup$

(c) $\cap$

(d) $\emptyset$

2. C_1, C_2 之各列有 3 种情形

① 兩行皆為 0 (type Z)

② 兩行皆為 1 (type X) (intersection)

③ 一行為 1, 一行為 0 (type Y)

令 $\begin{cases} \text{type X 有 } x \text{ 个 row} \\ \text{Y " } y \text{ " } \end{cases}$, 則 $\text{SIM}(C_1, C_2) = \frac{x}{x+y} = \frac{|C_1 \cap C_2|}{|C_1 \cup C_2|}$

假設執行一次 hash, 隨機排列各 row, 自上而下,
在 type Y 前先碰到 type X 之機率為 $\frac{x}{x+y}$, 此時
 $h(C_1) = h(C_2)$

若先碰到 type Y, 此時 $h(C_1) \neq h(C_2)$

故對任意 random permutation π ,

$$P(h_\pi(C_1) = h_\pi(C_2)) = \frac{x}{x+y} = \text{SIM}(C_1, C_2) \quad \times$$

3. Flajolet-Martin algorithm:

(1) hash function h st. h maps each of the N
Pick a
elements to at least $\log_2 N$ bits

(2) For each stream element a , 令 $r(a)$ 為 $h(a)$ 之
左端有幾個連續的 0

(3) 令 $R = \max_a r(a)$, 則 number of distinct element

的估計值 $= 2^R$

Why it works? 令 $h(a)$ 末端有 r 个 0, 若 1, 0 出現機率均為 $\frac{1}{2}$, 則 $P(m \text{ element 中找不到有 } r \text{ 个 0 的 element}) = (1 - \frac{1}{2^r})^m$

$$\Rightarrow (1 - 2^{-r})^m = (1 - 2^{-r})^{2^r m 2^{-r}} \underset{r \rightarrow \infty}{=} e^{-m 2^{-r}} \Rightarrow \begin{cases} 1, & \text{if } m \ll 2^r \\ 0, & \text{if } m \gg 2^r \end{cases}$$

$\Rightarrow$ if $m \ll 2^r$, then 找到 tail of length r 之 机率 $\rightarrow 0$
if $m \gg 2^r$, then 找到 tail of length r 之 机率 $\rightarrow 1$

$\Rightarrow$ 约有 2^r 个不同的 element

$\Rightarrow$ 约有 2^R 个

在此 stream 中

~~4. Break M into stripes. to avoid k -times scanning of M~~

~~$\Rightarrow$ scanning M and breaks it takes $|M|(1 + \epsilon)$~~

~~k scans of $r \Rightarrow (1+k)|r|$~~

4. $\therefore$ There's k block

$\therefore$ we need k scans of M and r

$\Rightarrow$ Cost per iteration: $k(|M| + |r|) + |r|$

$= k|M| + (k+1)|r|$

However, if we break M into stripes where each stripes contains only destination nodes in the corresponding block of r , then the cost reduce to $|M|(1 + \epsilon) + (k+1)|r|$, for some $\epsilon > 0$

~~✗~~

$$6(a) \text{ support } (\{1\}) = 4 \quad \{4\} = 8$$

$$\{2\} = 6$$

$$\{5\} = 6$$

$$\{3\} = 8$$

$$\{6\} = 4$$

$$\text{support } (\{1, 2\}) = 2 \quad \{2, 3\} = 3 \quad \{3, 5\} = 4$$

$$\{1, 3\} = 3$$

$$\{2, 4\} = 4$$

$$\{3, 6\} = 2$$

$$\{1, 4\} = 2$$

$$\{2, 5\} = 2$$

$$\{4, 5\} = 3$$

$$\{1, 5\} = 1$$

$$\{2, 6\} = 1$$

$$\{4, 6\} = 3$$

$$\{1, 6\} = 0$$

$$\{3, 4\} = 4$$

$$\{5, 6\} = 2$$

$$(b) (1, 2) \rightarrow 2$$

$$(2, 4) \rightarrow 8$$

$$(1, 6) \rightarrow 6$$

$$(1, 3) \rightarrow 3$$

$$(2, 5) \rightarrow 10$$

$$(2, 6) \rightarrow 1$$

$$(1, 4) \rightarrow 4$$

$$(3, 4) \rightarrow 1$$

$$(3, 6) \rightarrow 7$$

$$(1, 5) \rightarrow 5$$

$$(3, 5) \rightarrow 4$$

$$(4, 6) \rightarrow 2$$

$$(2, 3) \rightarrow 6$$

$$(4, 5) \rightarrow 9$$

$$(5, 6) \rightarrow 8$$

(c) 1, 2, 6, 4, 8 出現 2 次 + frequent buckets

(d) 沒 hash 到 frequent bucket 者 必非 frequent pair, 反之亦然

$\Rightarrow$ 能到 second pass 的有 (3, 4), (2, 6), (1, 2),

(4, 6), (2, 3), (1, 6), (1, 4), (3, 5), (2, 4), (5, 6)

7. false negative rate: all bands are not equal

$$20 \text{ bands, 5 rows} \Rightarrow (1 - (0.8)^5)^{20} \approx 0.0035$$

$$15 \text{ bands, 5 rows} \Rightarrow (1 - (0.8)^5)^{15} > 0.0035$$

false positive rate:

$$\begin{aligned} & 1 - (1 - 0.3^5)^2 \doteq 0.0474 \\ & 1 - (1 - 0.3^5)^5 \doteq 0.035 \end{aligned}$$

at least 1 band is equal

8.

∵ 共有 N bits

∵ 共需 $\log N$ 个 bucket. 設一个 bucket 需存 " n "

这个数, n 为 1 的权重 $\varnothing$

$$\Rightarrow \mathcal{O}(N) = \mathcal{O}(2^n) \Rightarrow \mathcal{O}(\log N) = (n \log 2) \Rightarrow \mathcal{O}(n) \leq \mathcal{O}(\log N)$$

$$\Rightarrow \text{共需 } \log N \cdot \mathcal{O}(\log N) = \mathcal{O}(\log^2 N) \text{ 个 bit}$$

(b). 

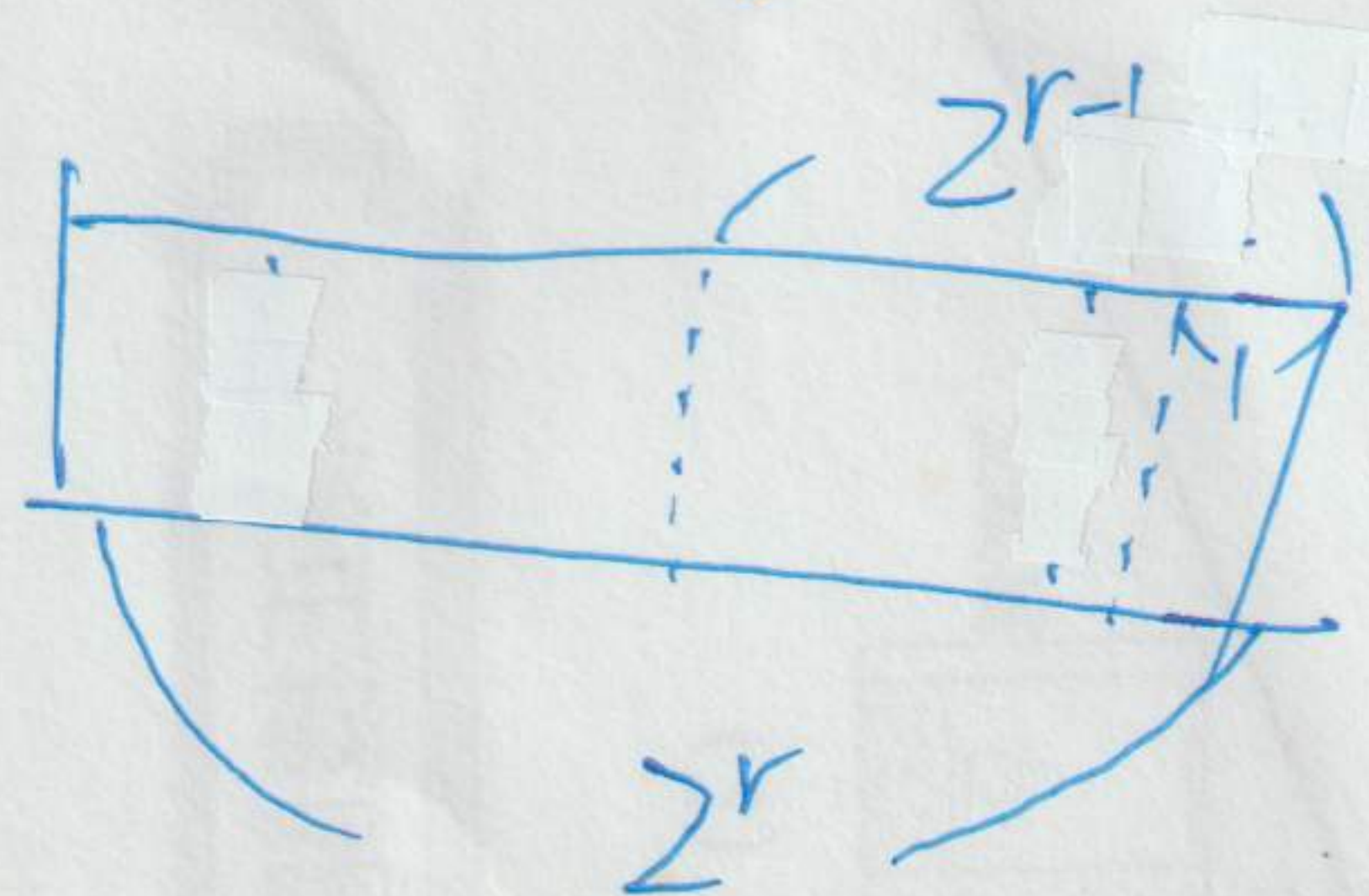
size 2^r 2^{r+1}

的 bucket

caused

$$\text{max error} = \frac{\text{max(error by size } 2^r \text{ 的 bucket)}}{\text{min(true sum)}}$$

$$= \frac{2^r - 2^{r-1} - 1}{1 + 2 + \dots + 2^{r-1} + 1} = \frac{2^{r-1} - 1}{2^r} = \frac{1}{2} - \frac{1}{2^r} < \frac{1}{2}$$



∴ 至少接收到 1 个 bit

要放进 size 2^r 的 bucket.

5. (a). Let $A_1, A_2, \dots, A_N$ be N advertisers.

Each has budget $B > N$. Suppose advertiser A_i bids on queries $q_1, \dots, q_i$ and no other queries, say Round i queries: bidders $A_i, \dots, A_N$

$$(b) \quad NB - B \left(\frac{1}{N} + \frac{1}{N-1} + \dots + \frac{1}{N-j+1} \right)$$

$$\stackrel{\uparrow}{=} B \left(N - \frac{N}{e} \right) = BN \left(1 - \frac{1}{e} \right)$$

Euler's approach

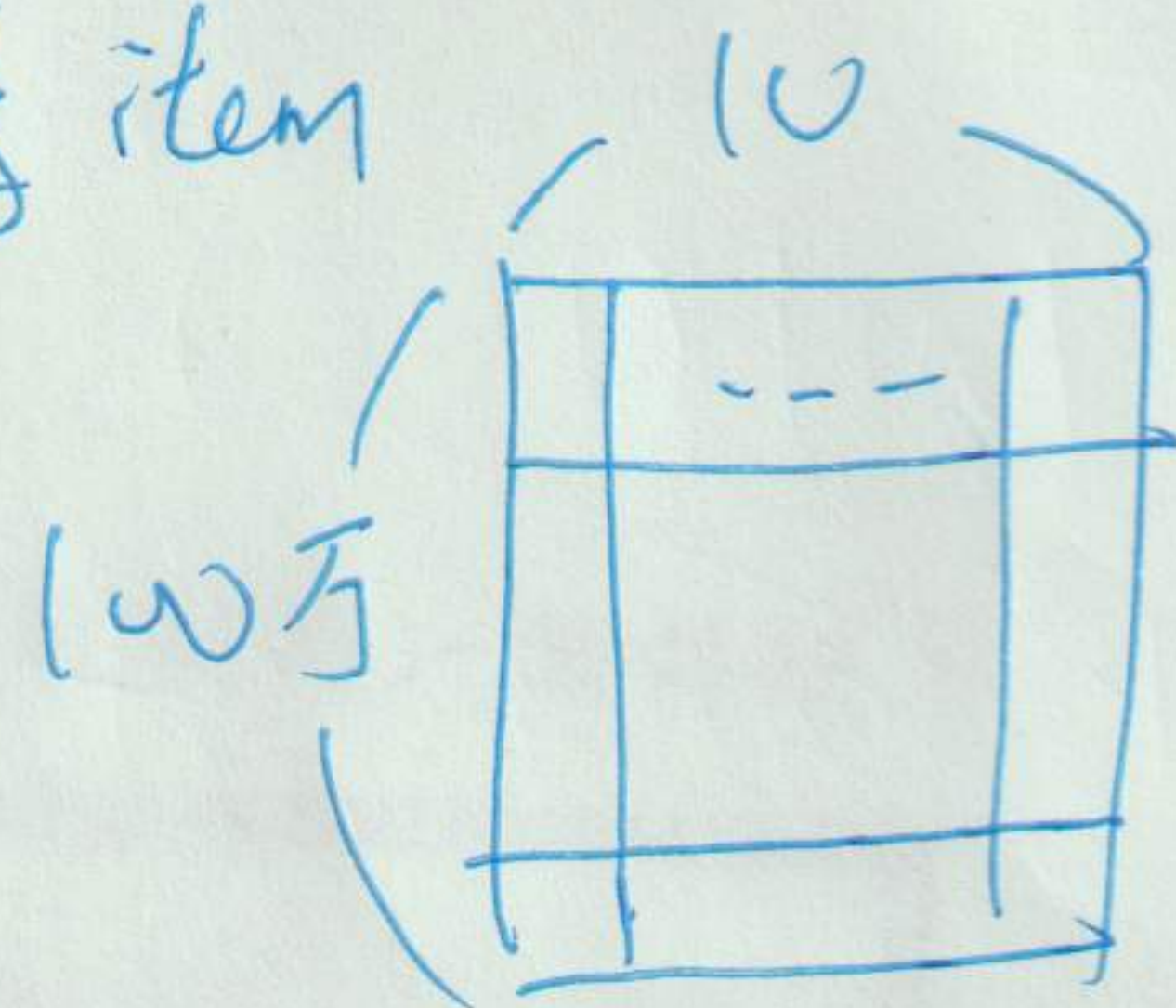
$$\text{competitive ratio} = \frac{BN(1 - \frac{1}{e})}{BN} = 1 - \frac{1}{e}$$

10. (a) maximum of frequent item: *

取理想下, 設前 10^5 basket 恰皆為 item pair $(a_1, \dots, a_{10})$, $10^6 \sim 2 \times 10^6$ 為 $(a_{11} \sim a_{20})$

依此類推, 有 $\frac{10^6}{10^5} \times 10 = 10$ 萬,

即全為 frequent item set



minimum of frequent item: 最極端下, 可只有 1 了 (obviously)

(b) 化簡問題: 有 10^5 basket, 各有 5 了 item,

每 了 item 為 $C_2^{10^6}$ 不同的 item pair 了 —

$\because C_2^{10^6} > 10^6 \therefore \min = 0$, 又 max 與 a 同理, $n = 10^5$

$$\text{max} = \frac{C_2^{10^6} \times 10^6}{10^5} = 45 \text{ 萬} *$$

兩 了 pair

12. (a) $aV = [0 \ 4 \ 5 \ 2 \ 0] \begin{bmatrix} 0.58 & 0 \\ 0.58 & 0 \\ 0.58 & 0 \\ 0 & 0.71 \\ 0 & 0.71 \end{bmatrix} = \begin{bmatrix} 5.220 & 1.420 \end{bmatrix}$

$\uparrow$ CS $\uparrow$ Mechanical engineering

(b) $\because 5.22 > 1.42$

$\therefore$ He prefers MADS, rather than Mechanics

13.

$\because \lambda$ is hyper parameter, W.L.O.G, $\hat{=}$

$$E = \sum_{(i,u)} (R_{iu} - g_i \cdot p_u^T)^2 + \frac{\lambda}{2} \left(\sum_u \|p_u\|_2^2 + \sum_i \|g_i\|_2^2 \right)$$

$$\Rightarrow \epsilon_{iu} = \frac{\partial E}{\partial R_{iu}} = 2(R_{iu} - g_i \cdot p_u^T)$$

又, 为了找 update equation, 计算梯度,

$$\begin{aligned} \nabla g_i &= -2(R_{iu} - g_i \cdot p_u^T) p_u^T + \lambda g_i \\ &= -(\epsilon_{iu} p_u^T - \lambda g_i) \end{aligned}$$

$$\Rightarrow \text{update equation for } g_i \leftarrow g_i - \mu \nabla g_i(R_{iu})$$

即 $g_i \leftarrow g_i + \mu(\epsilon_{iu} p_u^T - \lambda g_i)$

同理, $\nabla p_u^T = -2(R_{iu} - g_i \cdot p_u^T) g_i + \lambda p_u$

$$= -(\epsilon_{iu} g_i - \lambda p_u)$$

$$\Rightarrow p_u^T \leftarrow p_u^T - \mu_2 \nabla p_u^T(R_{iu})$$

$$\Rightarrow p_u^T \leftarrow p_u^T + \mu_2 (\epsilon_{xi} g_i - \lambda p_u^T)$$

$$14. (a). y = \beta M \left(\frac{\beta y}{M} + \frac{1-\beta}{N} \right) + \frac{1-\beta}{N} + x$$

$$= \beta^2 y + \frac{M}{N} \beta(1-\beta) + \frac{1-\beta}{N} + x$$

$$\Rightarrow (1-\beta^2)y = \frac{M}{N} \beta(1-\beta) + x \quad \text{small, ignore}$$

$$\Rightarrow y = \frac{\beta}{1-\beta^2} \cdot \frac{M}{N} + \frac{\beta^2 y}{1-\beta^2} + \frac{x}{1-\beta^2} \text{ two small.}$$

(b). $\rightarrow$ 題目有錯?

$$\text{左: } r_{p_0} = \sum_{i \rightarrow p_0} \beta \frac{r_i}{d_i} + (1-\beta) \frac{1}{N} = \frac{\beta}{2} \left(\sum_{i=1}^k r_{p_i} + \sum_{i=1}^m r_{g_i} \right) +$$

$$r_{g_0} = \sum_{i \rightarrow g_0} \beta \frac{r_i}{d_i} + (1-\beta) \frac{1}{N} = \frac{\beta}{2} \left(\sum_{n=1}^k r_{p_i} + \sum_{i=1}^m r_{g_i} \right) +$$

$$\text{右: } r_{p_0} = \sum_{i \rightarrow p_0} \beta \frac{r_i}{d_i} + (1-\beta) \frac{1}{N} = \beta \sum_{n=1}^k r_{p_i} + \frac{\beta}{1} r_{g_0} + (1-\beta)$$

$$\text{同理 } r_{g_0} = \beta \sum_{n=1}^k r_{p_i} + \frac{\beta}{1} r_{p_0} + (1-\beta) \frac{1}{N} \quad \uparrow \text{rel}$$

$$\Rightarrow r_{p_0} = \left(\beta + \beta^2 \right) \sum_{n=1}^k r_{p_i} + (1-\beta^2) \frac{1}{N} + \beta^2 r_{p_0}$$

$$\Rightarrow r_{p_0} = \frac{\beta + \beta^2}{1-\beta^2} \sum_{n=1}^k r_{p_i} + \frac{1}{N} = \frac{\beta}{1-\beta} \sum_{n=1}^k r_{p_i} + \frac{1}{N}$$

$$\text{同理 } r_{g_0} =$$

$$\frac{\beta}{1-\beta} \sum_{n=1}^m r_{g_i} + \frac{1}{N}$$

Take $\beta = \alpha t$, $\sum_c r_{p_c} = P$, $\sum_c r_{g_c} = Q$

左圖: $r_{p_0} = \frac{2}{5}(P+Q) + \frac{1}{5}\frac{1}{N} = r_{g_0}$

右圖: $r_{p_0} = 4P + \frac{1}{N}$, $r_{g_0} = 4Q + \frac{1}{N}$

$\Rightarrow$ If $P \approx Q$, 右圖較佳 ~~✗~~

11. (a) $yzyy \rightarrow$ Yes, optimum only yields \$3
opt = $\overline{A}B\overline{A}$

(b) $xyzx \rightarrow$ Yes, optimum = 4\$
opt: $\overline{A}A\overline{B}B$
or $BABA$

(c) $xyyy \rightarrow$ No, If the first x is assigned to A , then the yield is only \$2 but \$3 = optimum

(d) $xyxz \rightarrow$ No, optimum = 4 ($AABB$ for example),
but if both x are assigned to B
 z can't be satisfied ~~✗~~

9. (a). Since, If the item in S we surely
output it, if not, we still output it

(b). Suppose we have x targets and
 y darts. Any darts is equally likely
to hit the target.

$\Rightarrow$ The probability that none of the y darts will hit the given target is $\left(\frac{x-1}{x}\right)^y = \left(\frac{x-1}{x}\right)^{x \cdot \frac{y}{x}}$

$$= \left(1 - \frac{1}{x}\right)^{x \cdot \frac{y}{x}} \approx e^{-\frac{y}{x}}$$

$\therefore$ there are km darts

$\therefore$ fraction in 1's is $1 - e^{-\frac{km}{n}}$

$\Rightarrow$ FP rate $= \left(1 - e^{-\frac{km}{n}}\right)^k$ since there are k indep. hash functions. and

$$\text{Optimal } k = \frac{n}{m} \ln 2. \quad \#$$